



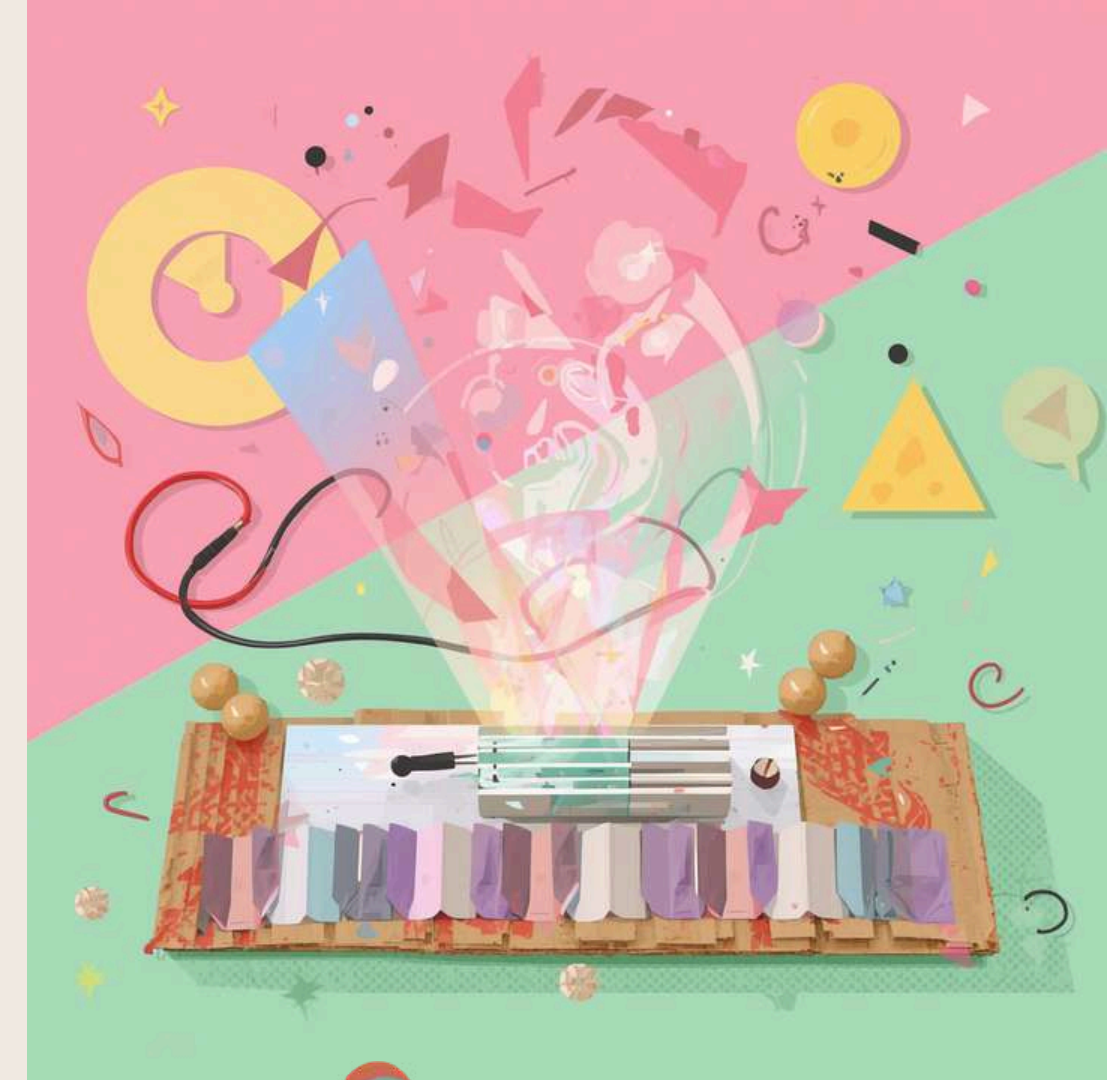
CIRCUITS: BUZZER MUSICAL KEYBOARD AND MORE!



WOW! LISTEN TO THIS KEYBOARD

DEKHO! A KEYBOARD MADE WITH CIRCUITS!

A keyboard made with circuits and a buzzer — Sa Re Ga Ma sounds! Press the foil strips to complete the circuit and hear musical notes come alive!



An abstract illustration featuring various colorful shapes like circles, triangles, and stars in shades of red, orange, yellow, green, and blue. These shapes are interconnected by thin, winding lines, creating a network-like pattern that suggests a circuit or a web of connections.

WHY DO CIRCUITS MATTER?

CIRCUITS ARE EVERYWHERE!

What is a circuit? It's a path that lets electricity flow to make things work! We use circuits every single day without even thinking about it.

Examples: Diwali lights, ghar ki ghanti (doorbell), auto-rickshaw horn, TV remote. Question for you: "Acha! Can you think of other things with circuits at home?"



WHAT IS AN ELECTRIC CIRCUIT?

CIRCUIT = PATH FOR ELECTRICITY TO FLOW!

Open circuit = electricity stops (like a broken road!)

Closed circuit = electricity flows (complete path!)



CONDUCTORS AND INSULATORS

WHAT MATERIALS LET ELECTRICITY FLOW?

Conductors: Let electricity flow easily!

- Metal, aluminum foil, copper wires
- Example: Auto-rickshaw horn wires

Insulators: Stop electricity from flowing!

- Plastic, rubber, wood
- Example: Plastic handles on kitchen tools



QUICK CHECK: CIRCUITS AT HOME

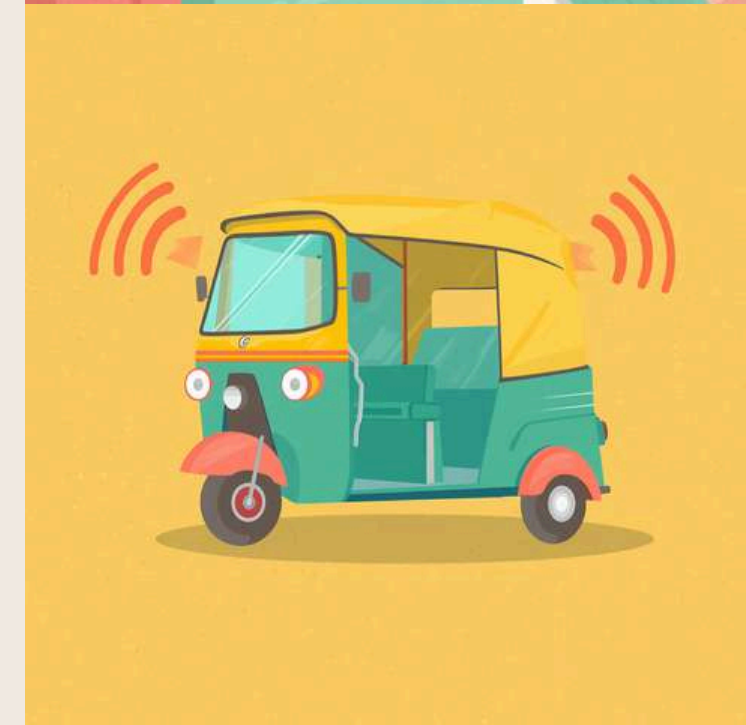
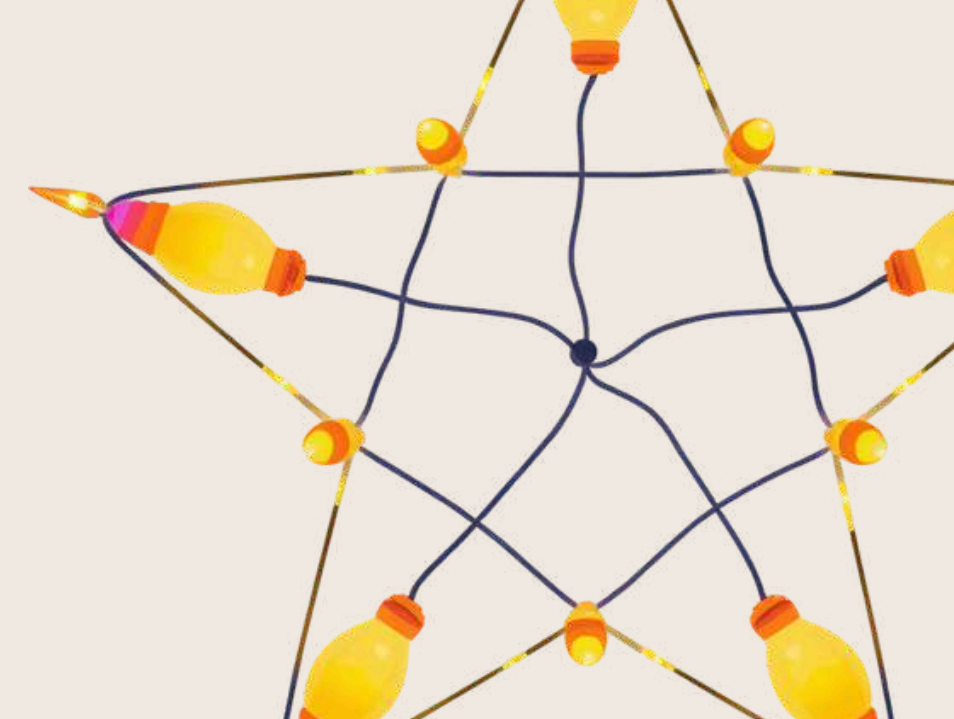
DEKHO! CIRCUITS ARE EVERYWHERE AROUND US. CAN YOU SPOT THEM?

Diwali Lights: When the circuit is closed, all the tiny bulbs light up and make your home sparkle!

Ghar ki Ghanti: Press the doorbell button and the circuit closes — ding dong! Someone's at the door!

Auto-Rickshaw Horn: The driver presses the horn, the circuit completes, and PON PON! Everyone moves aside!

TV Remote: Press a button and tiny circuits inside send signals to change channels. Magic!





WHAT IS DIFFUSION?

PARTICLES ON THE MOVE!

Diffusion = particles moving from where there are many to where there are few. It's like when something spreads out to fill a space evenly!

Dekho! Have you smelled agarbatti or chai aroma spreading around the room? That's diffusion in action — the smell particles travel from the source to your nose!



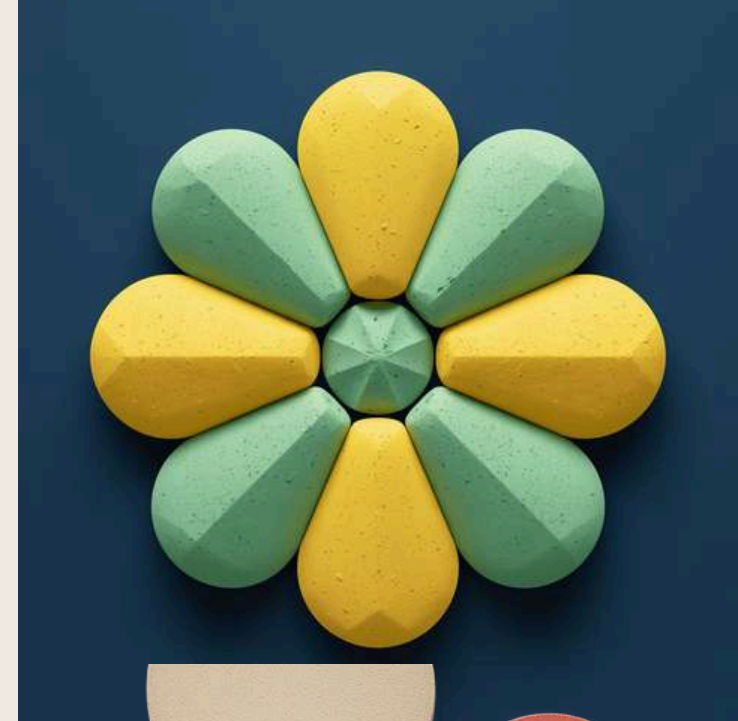
DIFFUSION IN ACTION!

AGARBATTI SMOKE SPREADING

Dekho! When you light an agarbatti, the smoke particles spread from where there are many to where there are few. That's why the whole room smells nice!

GULAL FLYING DURING HOLI

When you throw gulal powder in the air, the tiny color particles spread everywhere through diffusion. That's how Holi colors reach all your friends!





MORE DIFFUSION EXAMPLES

SMELLS & COLORS THAT SPREAD!



🍵 Chai aroma spreading from the kitchen

🚰 Smell at a petrol pump reaching your nose

🎨 Holi colors mixing in water buckets

🌸 Perfume spreading in a room

🔍 Cooking smells traveling through the house

❓ Can you think of other smells or colors that spread like this?



WHAT ARE COLOURS & PIGMENTS?

UNDERSTANDING LIGHT, ABSORPTION & REFLECTION

Pigments are special substances that give things their color! They work by absorbing some colors of light and reflecting others back to our eyes. That's how we see different colors all around us!

Indian Examples: Haldi (turmeric) looks yellow because it reflects yellow light. Mehndi gives greenish-brown color. Primary pigments are cyan, magenta, and yellow — mixing them creates new colors!





INDIAN COLOURS IN EVERYDAY LIFE!

PIGMENTS ALL AROUND US

Holi gulal powders – bright reds, pinks, yellows!

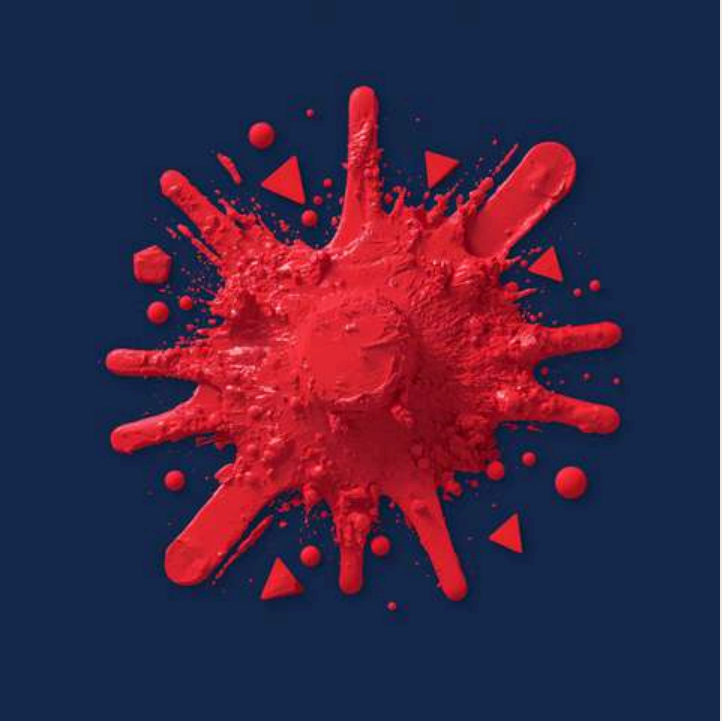
Rangoli powders – vibrant festival decorations

Saree dyeing – beautiful fabric colors

Indian flag – saffron, white, green & blue chakra

Mehndi – natural greenish-brown pigment





HOW PIGMENTS MIX



RED + YELLOW = ORANGE!

When you mix red gulal powder with yellow gulal powder, the pigments combine to create a beautiful orange color! Try it during Holi or with paint.



BLUE + YELLOW = GREEN!

Mix blue and yellow gulal or paint together and watch the magic happen — you get green! This is called subtractive color mixing.



SAFETY FIRST!

IMPORTANT RULES TO REMEMBER

- Always use batteries under 9V only
 - Never use wall sockets — they are dangerous!
- Always do experiments with a parent or teacher
 - Keep water away from circuits
 - Handle wires and batteries carefully



LET'S BUILD OUR BUZZER KEYBOARD!

READY TO MAKE MUSIC WITH CIRCUITS?

Materials needed: 9V battery, piezo buzzer, aluminum foil, cardboard, scissors, and tape.

Gather everything and let's create your own musical instrument!

STEP 1 - PREPARE THE BASE

CUT THE CARDBOARD BASE

Take a rectangular piece of cardboard — this will be the base of your buzzer keyboard. Make sure it's big enough to fit 5 keys side by side. Ask an adult to help you cut it to the right size!

ADD THE FOIL KEYS

Cut 5 strips of aluminum foil and tape them onto the cardboard, spaced out like piano keys. Leave a small gap between each strip — these will be your musical keys! Make sure each foil strip is smooth and flat.



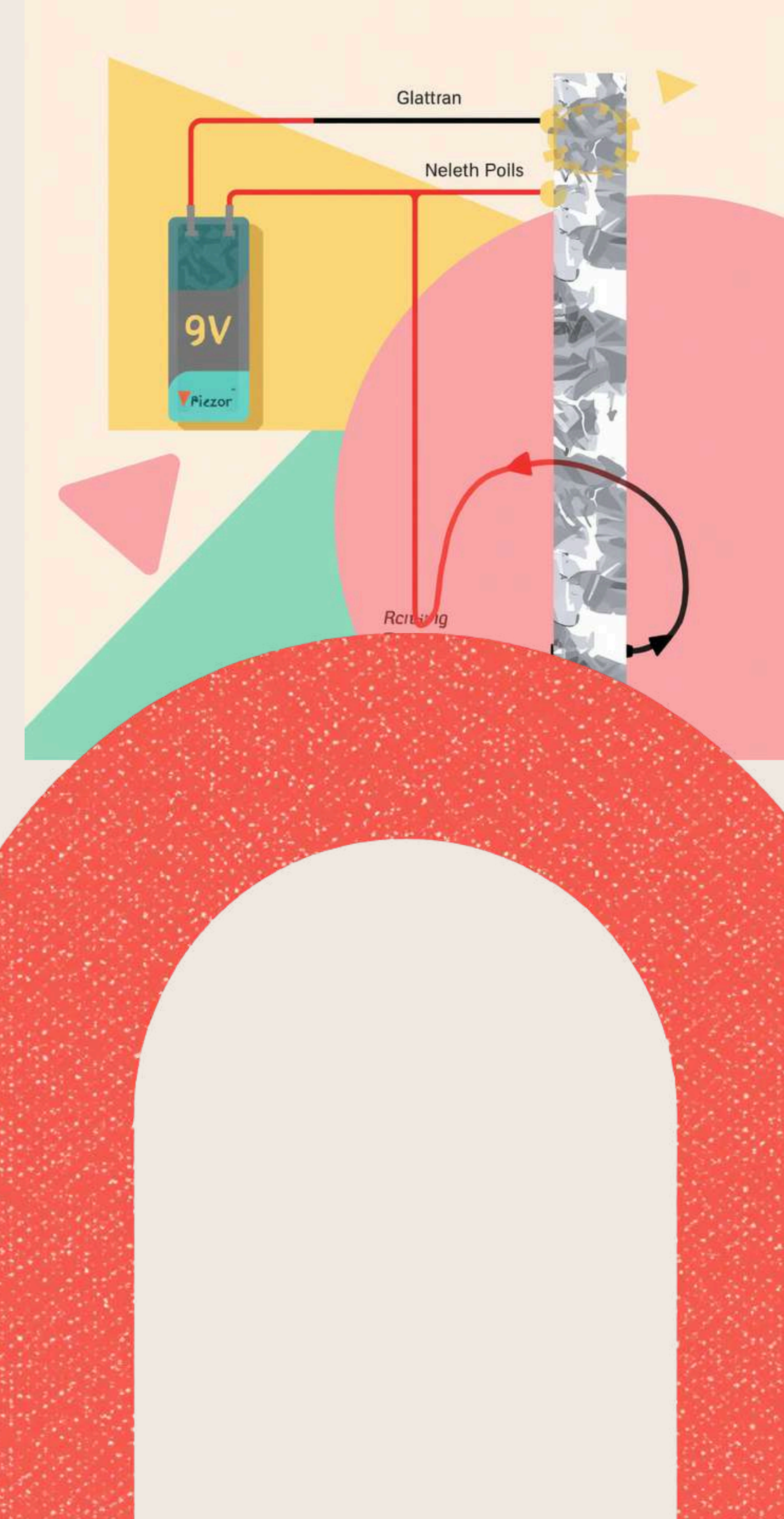
STEP 2 - CONNECT THE BATTERY AND BUZZER

RED WIRE CONNECTION

Attach the red wire from the battery's positive (+) terminal to one side of the piezo buzzer. Remember: Red = Positive!

BLACK WIRE CONNECTION

Attach the black wire from the battery's negative (-) terminal to the aluminum foil strip side. Black = Negative! This creates your circuit path.



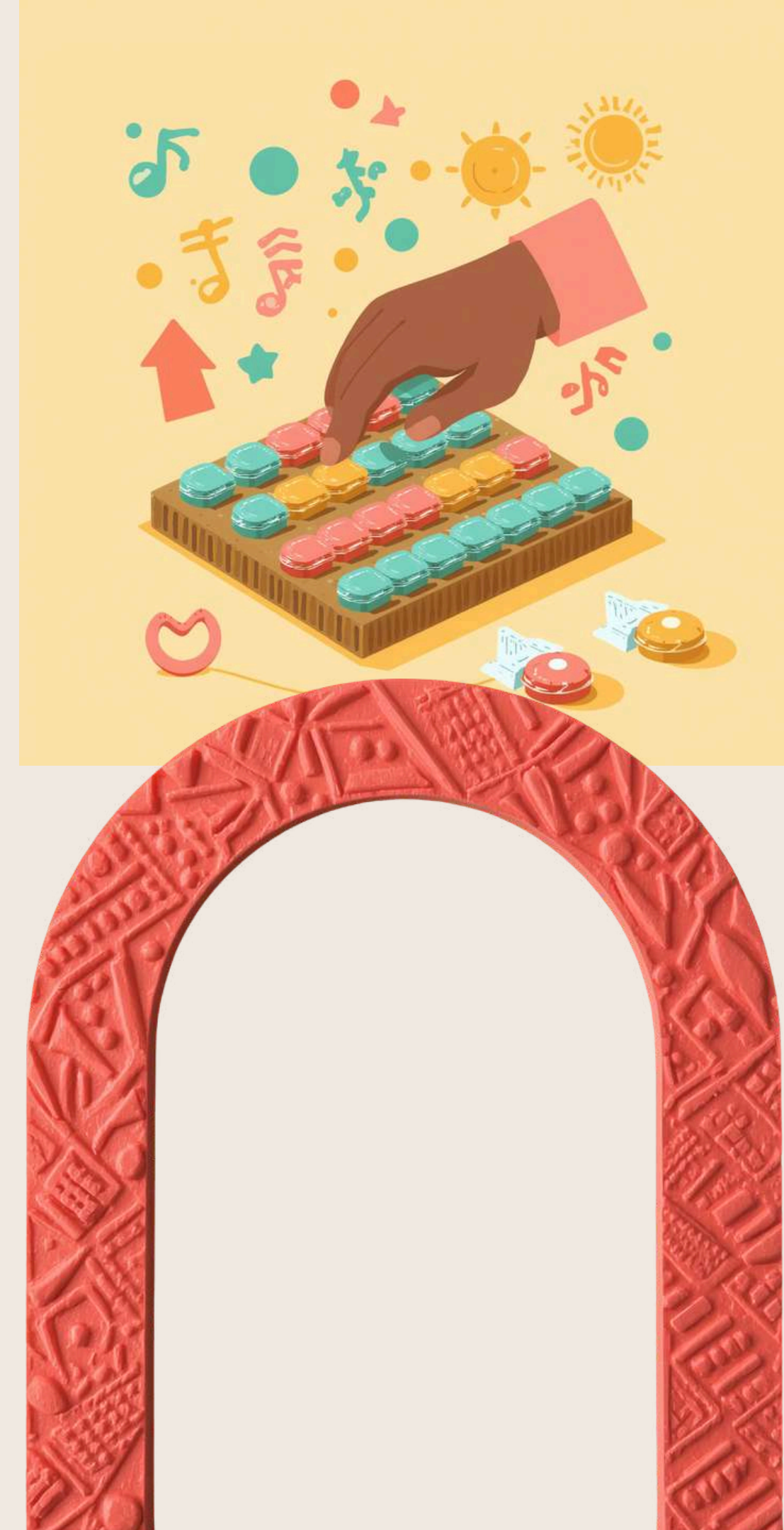
STEP 3 - COMPLETE THE CIRCUIT!

PRESS THE FOIL KEYS

Gently press each aluminum foil key with your finger. This connects the battery negative to the buzzer negative, completing the circuit. When the circuit closes, electricity flows and the buzzer makes a sound!

LISTEN TO THE MAGIC!

Each key can make a different sound! Try pressing different keys to hear the buzzer. When you lift your finger, the circuit opens and the sound stops. Bahut maza! You control the music!



STEP 4 - PLAY YOUR SA RE GA MA!

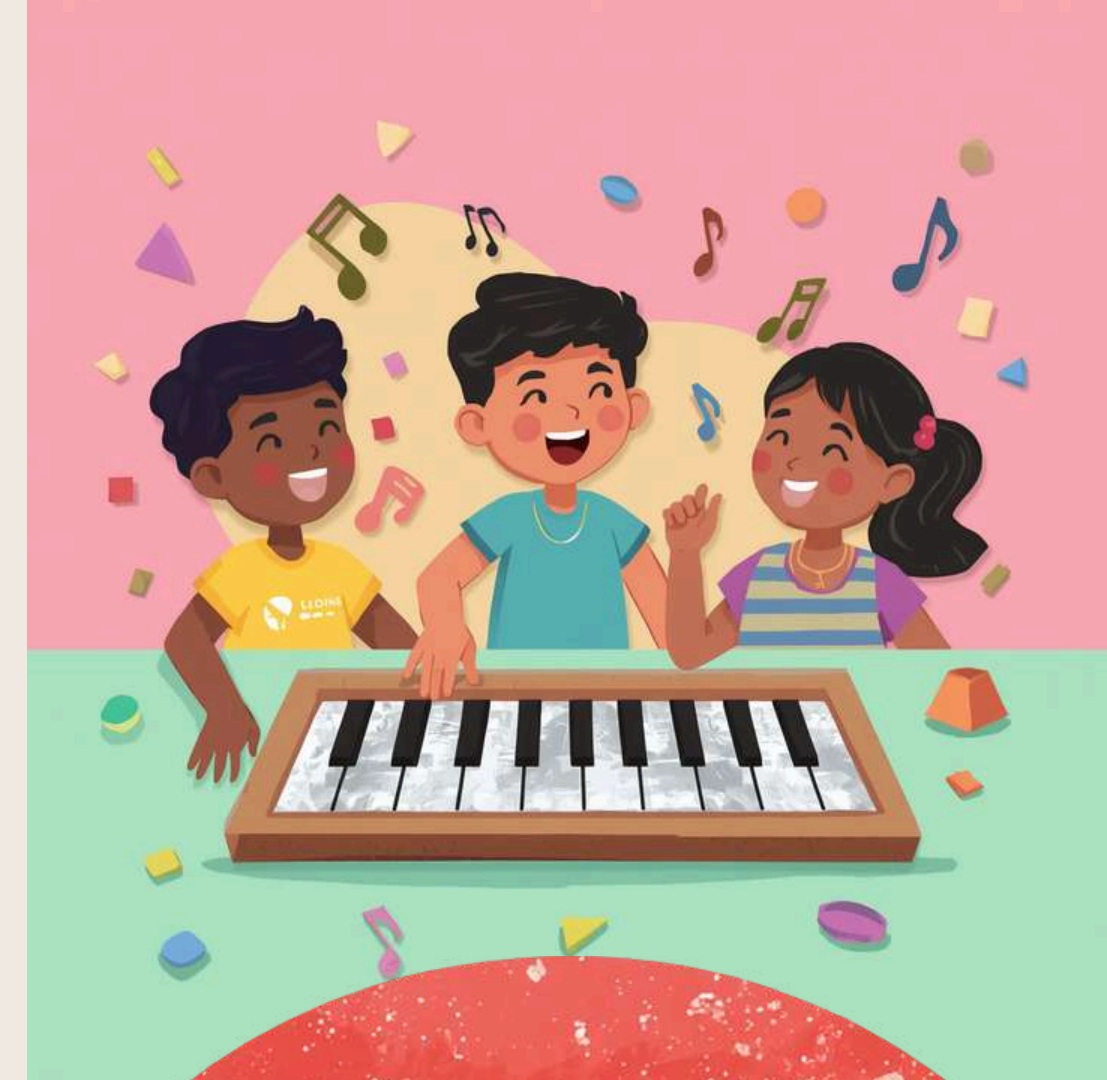
BAHUT BADIYA!

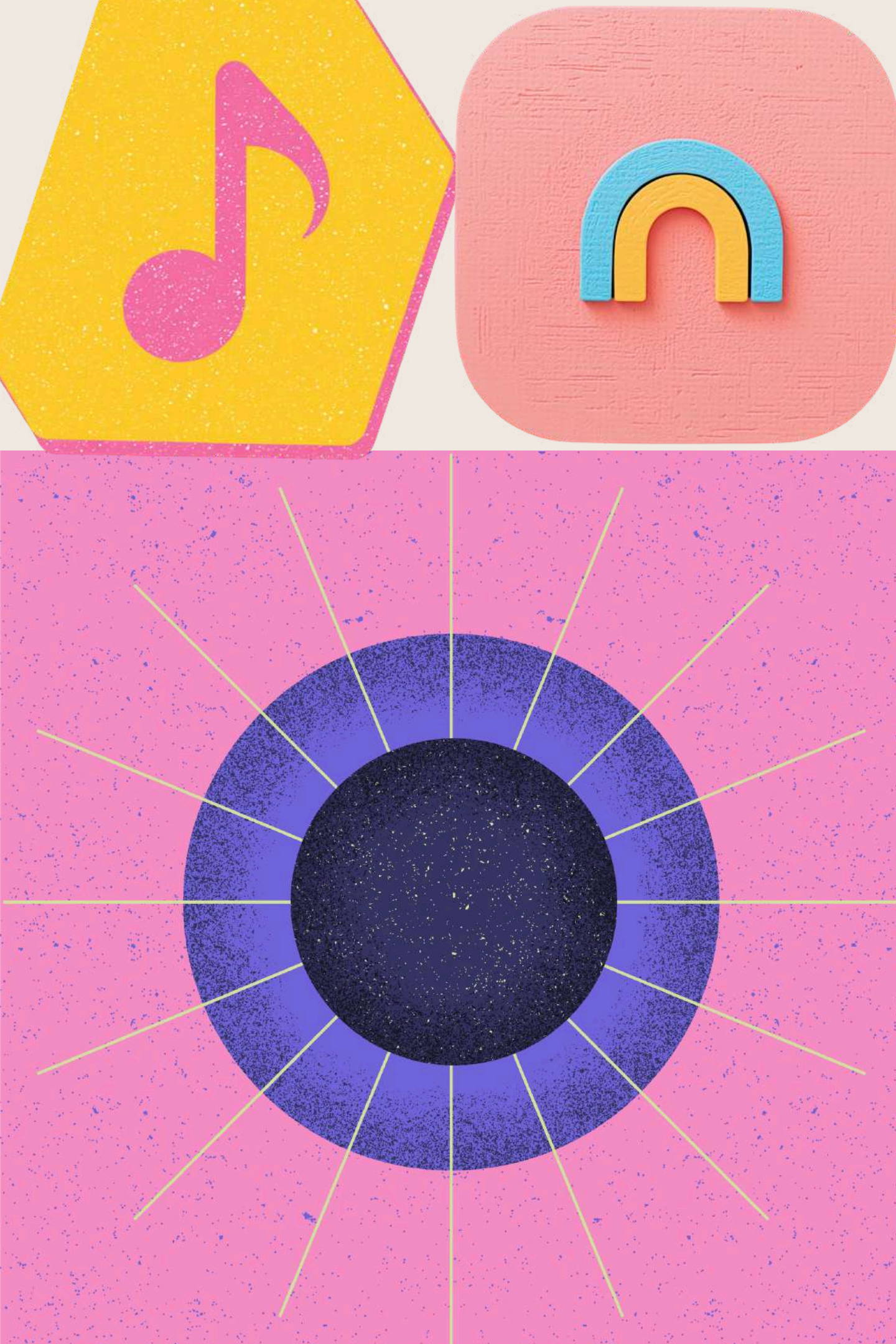
You made your own musical instrument! Press each foil key and listen to the buzzer make different sounds. Try playing Sa Re Ga Ma Pa on your keyboard!

YOU DID IT!

Share your buzzer keyboard with friends and family. Can you play a simple tune?

Keep experimenting with different rhythms and patterns!





WHAT DID WE LEARN TODAY?

CIRCUITS POWER EVERYTHING!

Electric circuits let electricity flow in a complete path to make things work — from Diwali lights to doorbells to your buzzer keyboard!

DIFFUSION & PIGMENTS

Diffusion spreads smells and colors all around us (like chai aroma!). Pigments give colors by absorbing and reflecting light. And YOU built a buzzer keyboard that plays Sa Re Ga Ma!

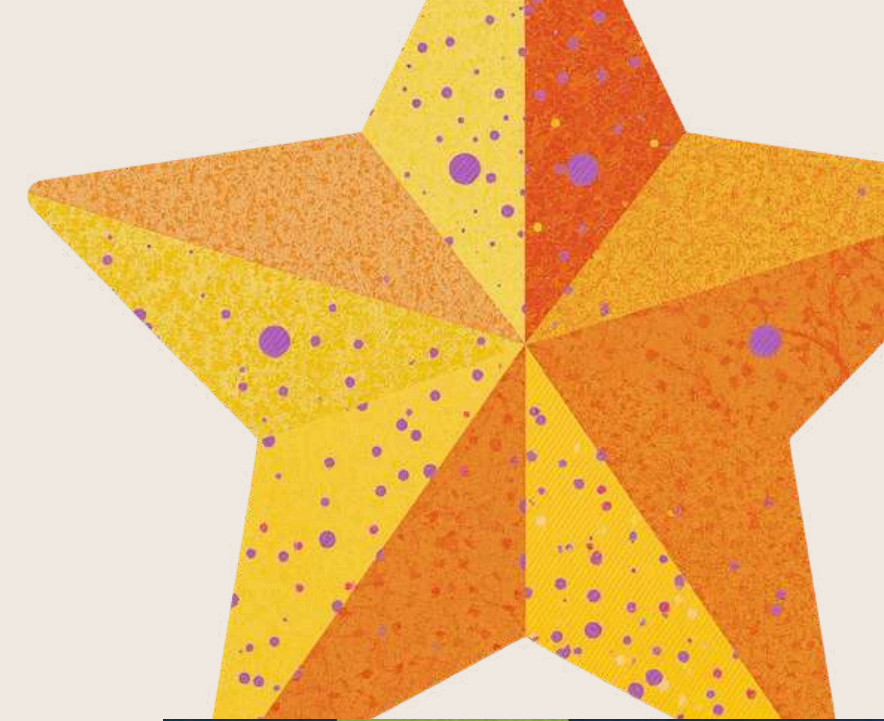
3-2-1 REFLECTION TIME

SOCHIYE AUR LIKHIYE! THINK ABOUT WHAT YOU LEARNED TODAY.

3 Things I Liked: Write three things you enjoyed most in today's science session about circuits, diffusion, and colors!

2 New Words I Learned: Write two new science words you learned today (hint: pigment, diffusion, circuit, conductor, insulator)

1 Thing I Want to Try at Home: Write one experiment or activity from today that you want to try again with your family!





**THANK YOU & KEEP
EXPLORING!**